

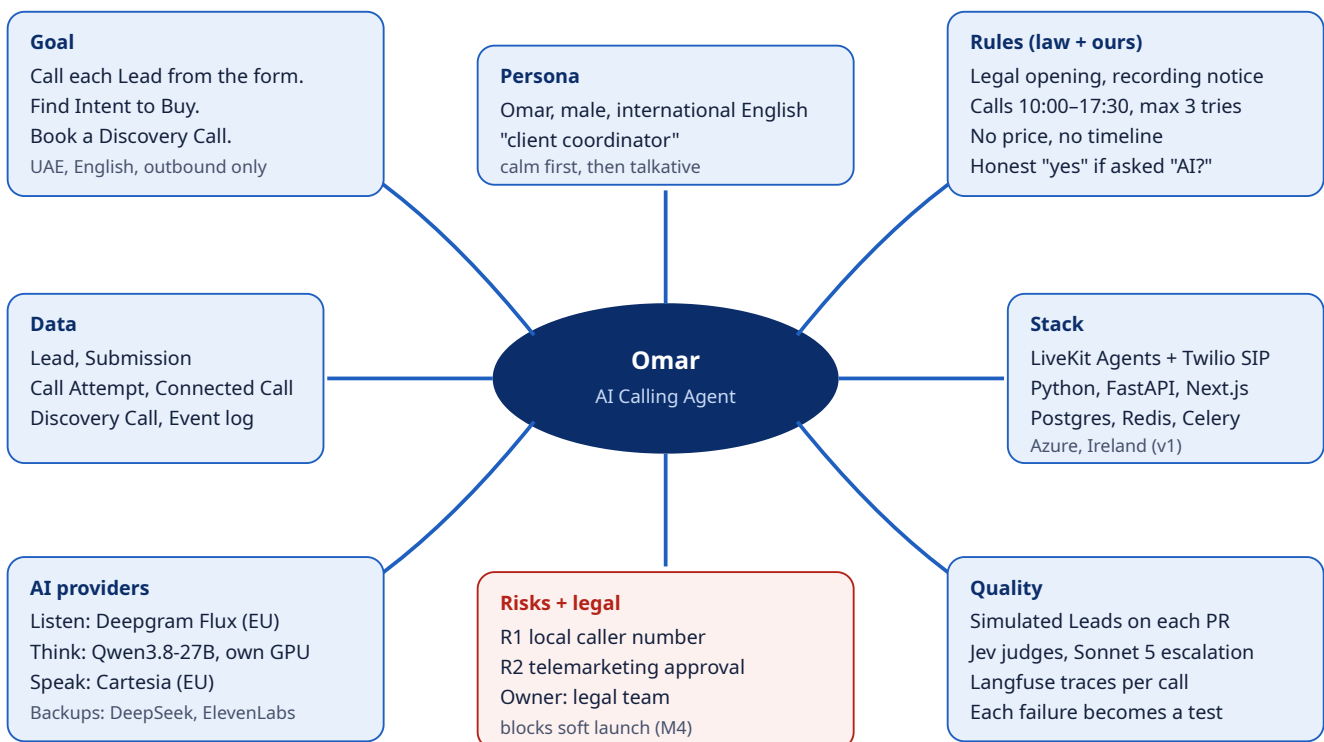
AI Calling Agent: Engineer Guide

The complete picture of "Omar": goal, rules, architecture, plan and risks

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3 October 2026 · Version 1.0

Read this first. This guide gives you the full project in short sections. Each section has a diagram or a chart. The source of each decision is a ticket in `.scratch/calling-agent/issues/`. The glossary is in `CONTEXT.md`. Use the words from the glossary. Do not use other words for the same things.

0 The project on one page



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1 Goal

Hoplon & Co gets Leads from the form on hoplonco.com. The Calling Agent (Omar) calls each Lead. Omar gives the information that the Lead asked for. Omar finds Intent to Buy. Then Omar books a Discovery Call with the Manager. This project is also a portfolio project. Each decision must be clear and explained.

Scope in three columns

v1 (this plan)

- Outbound calls only
- English only
- UAE Leads only
- Twilio + LiveKit, Azure Ireland
- Excel and Gmail ingestion
- Dashboard, admin only

v2 (later)

- All parts in Azure UAE North
- e&/du SIP trunk
- Self-hosted STT and TTS
- Temporal for the Lead journey

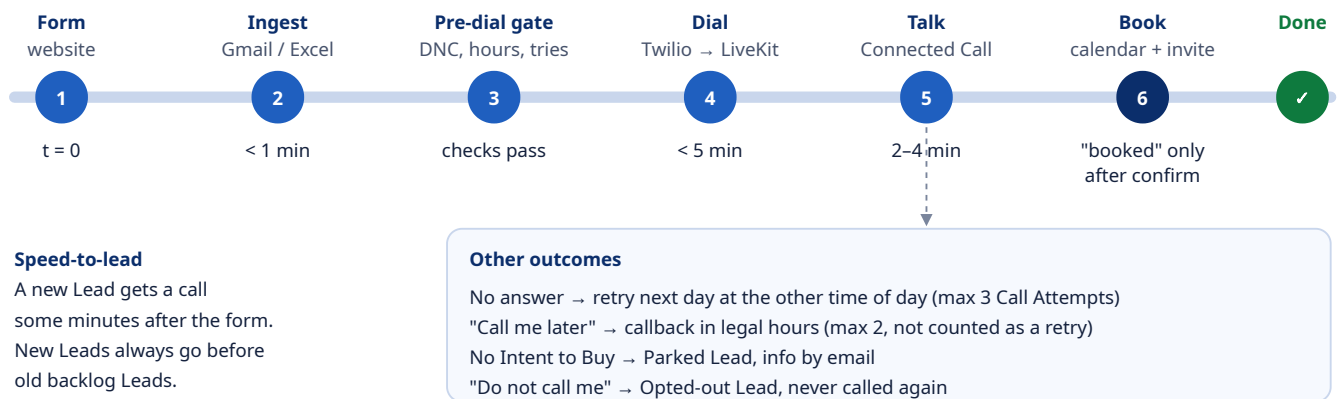
Out of scope

- CRM updates
- Arabic or other languages
- Inbound calls
- Product for clients
- WhatsApp (calls are blocked in UAE)
- Live transfer to the Manager

1.1 Glossary (from CONTEXT.md)

Term	Meaning	Do not say
Lead	A person who showed interest through our web form. Not qualified yet.	prospect, contact
Submission	One filled-in web form. One Lead can have many.	enquiry, form entry
Calling Agent	The AI system that calls Leads and talks by voice. Persona name: Omar.	bot, dialer
Call Attempt	Any outbound dial to a Lead. The Lead can answer or not.	dial
Connected Call	A Call Attempt with a live talk. It tries to find Intent to Buy and book a Discovery Call.	sales call, cold call
Intent to Buy	The Lead says "yes", asks about price, or asks to talk to a person.	BANT, score
Qualified Lead	A Lead who showed Intent to Buy on a Connected Call.	—
Parked Lead	No Intent to Buy, but the Lead did not reject contact.	dead lead
Opted-out Lead	The Lead rejected contact. We never call again.	DNC lead
Unreachable Lead	The Lead did not answer any permitted Call Attempt.	dead number
Discovery Call	A 30-minute meeting with the Manager. Budget and timeline are discussed only here.	demo, appointment
Manager	The human who hosts Discovery Calls. Omar says "my manager".	"a human"

2 The Lead journey



2.1 The Manager's day

Morning

Open the dashboard Home. See new bookings, flags and cost.

Before a meeting

Read the prep brief: summary, Intent to Buy evidence, transcript link.

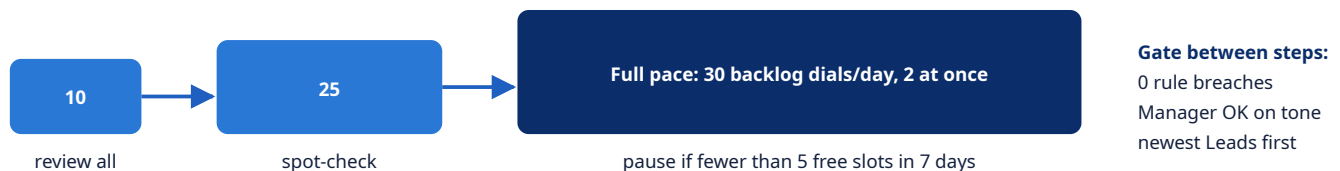
After a meeting

Click Attended, No-show, Won or Lost. This measures quality.

Weekly

Review flagged calls. Label transcripts. Start the next backlog batch.

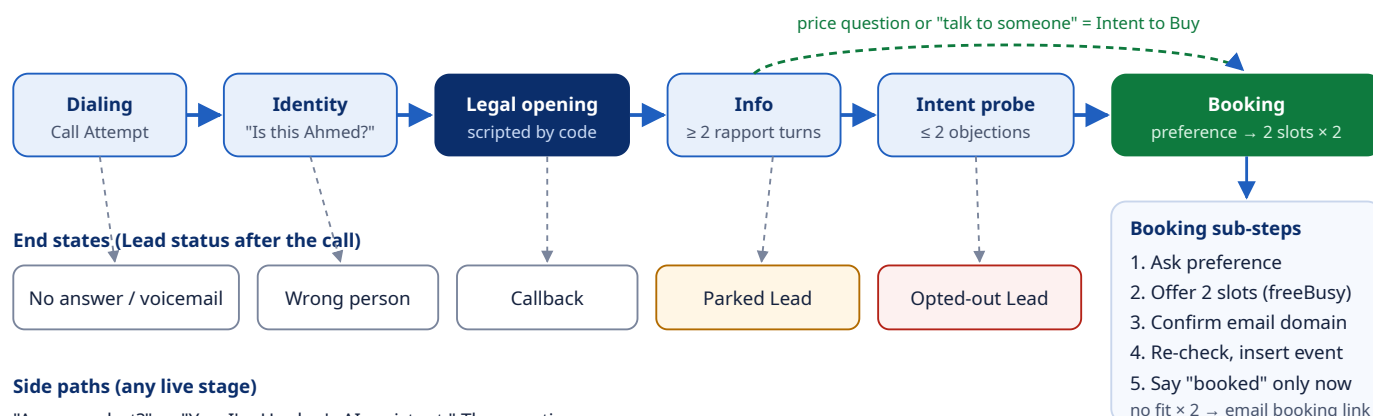
2.2 Backlog plan and soft launch



Old Leads (more than 30 days) hear a different opening: "You contacted us in March about an app. Are you still interested?" Non-UAE numbers are skipped and flagged.

Code owns the call state machine (ADR 0001). The LLM writes the words inside each stage. The LLM never decides alone to skip a rule. A clickable prototype is in [.scratch/calling-agent/prototypes/PROTOTYPE-call-flow.html](#).

3 Stages of a Connected Call



Side paths (any live stage)

"Are you a bot?" → "Yes, I'm Hoplon's AI assistant." Then continue.

Prompt injection → light laugh, "Nice try!", back to the topic.

Arabic only → "My manager speaks English. OK?" No → Parked (needs Arabic).

"I don't want recording" → end call, email info → Parked (declined recording).

"How much?" / "How long?" → "We adjust to your needs. My manager covers that."

Guards enforced by code (never by prompt only)

- Legal order: company + reason → recording → "continue?"
- No pitch before the Lead agrees to continue
- No budget or timeline numbers
- No "booked" before calendar confirms · max 10 min

3.1 The opening (legal order)

"Hi, is this {first name}?" → "Hi {first name}, this is **Omar from Hoplon and Co.** You filled in a form on our website about {topic}, so I'm calling with the information you asked for. **Just so you know, this call is recorded.** Is now a good time for two minutes?"

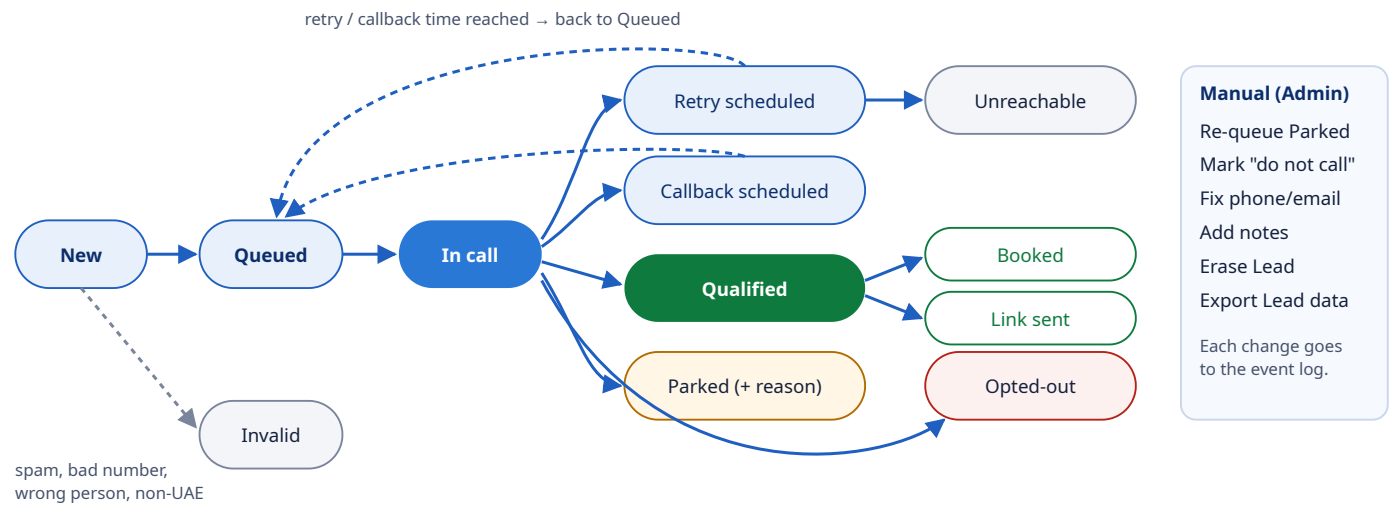
Lines in the prototype are samples. The real agent writes natural words inside each stage. Persona values are configuration, not code.

3.2 Objections and answers (examples)

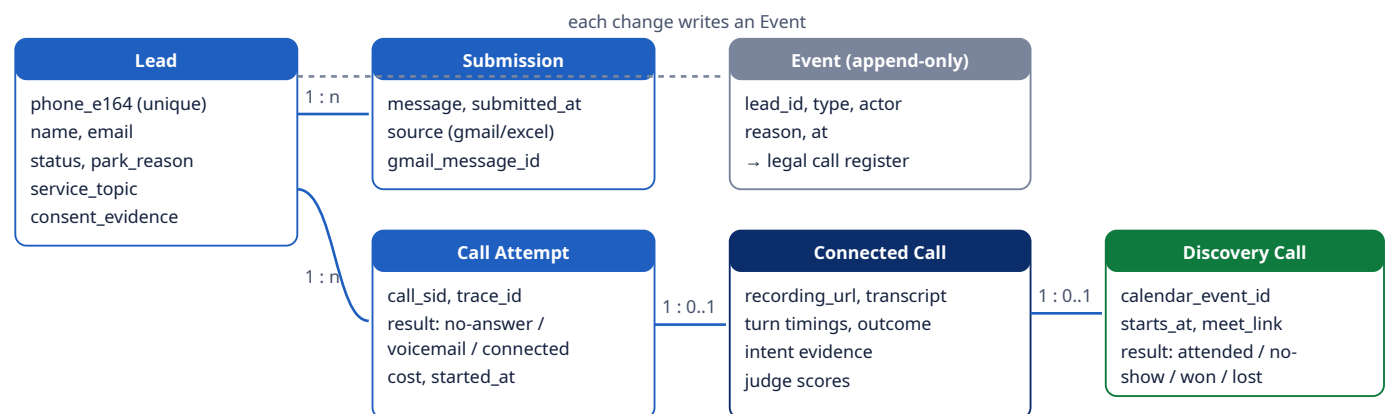
The Lead says	Omar does	Result
"Just send me an email."	Agrees. Says a short call with the Manager saves time. Offers 30 minutes this week. Max 2 tries.	3rd time → Parked
"How much will it cost?"	"It depends on your needs. We can adjust the budget. My manager covers that." Moves to booking.	Intent to Buy
"Can I talk to someone?"	"Of course. Let me connect you with my manager." Moves to booking.	Intent to Buy
"I'm busy, call me later."	Asks for a time. Agrees only inside the calling window.	Callback
"Just browsing, maybe next year."	Offers info by email. Ends the call warmly.	Parked
"Not interested. Don't call me."	"Understood, I won't call again." Ends the call.	Opted-out

4 Lead statuses

The Lead status follows the Lead across all calls. The call state machine follows one call only. Do not mix them.

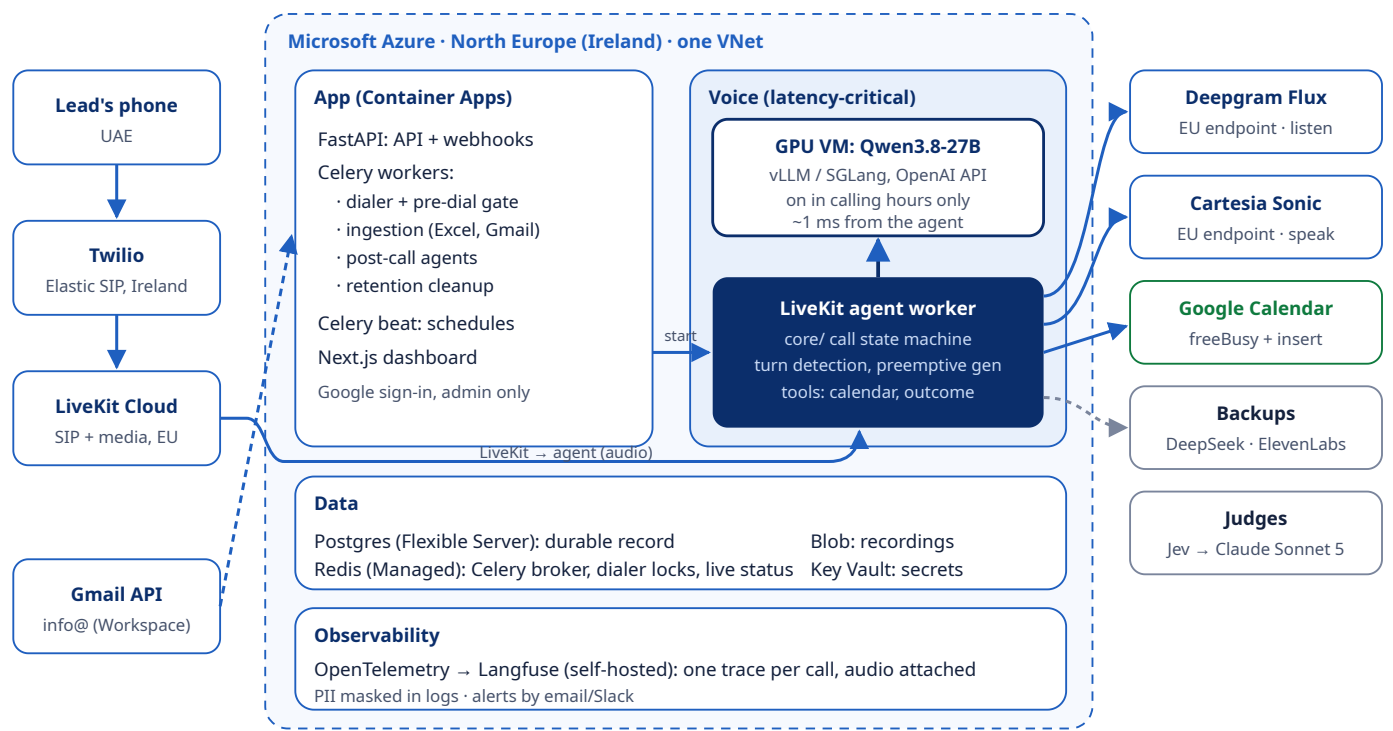


4.1 Data model (simplified)



Retention: recordings and transcripts are kept 12 months by default (configurable). A cleanup job deletes old data. "Erase Lead" removes personal data and keeps an anonymous record.

5 System view (v1)



What we build, and what we use

We build	We use
Call state machine, guards	LiveKit Agents (open source)
Domain model, data, dialer	LiveKit Cloud SIP + media
Ingestion, dashboard	Twilio Elastic SIP Trunking
Eval harness, judges setup	Deepgram, Cartesia, Qwen weights

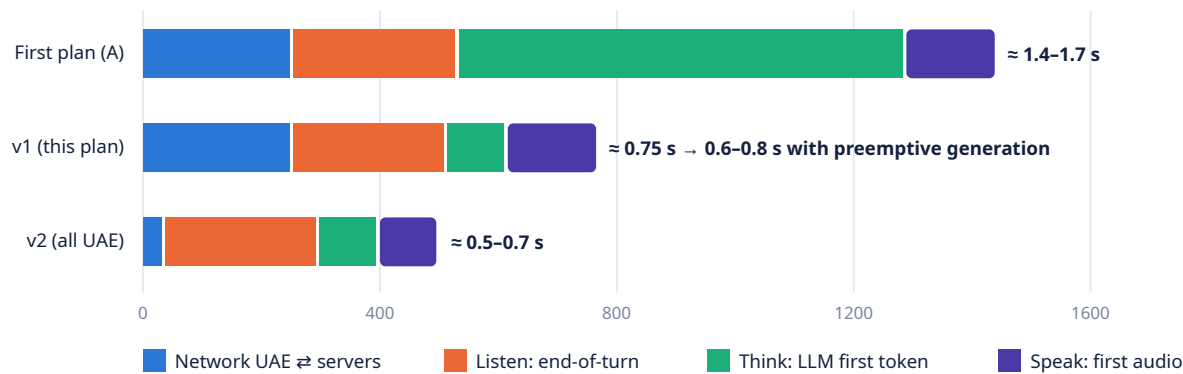
ADR 0002 records why we use LiveKit and do not hand-write the voice loop.

Repo layout (monorepo, uv workspace)

Folder	Contents
core/	Domain, state machine, ports. No I/O. No dependencies.
voice/	LiveKit agent worker
app/	FastAPI, Celery, dialer, ingestion
web/	Next.js dashboard (UI only)
evals/	YAML scenarios, harness, judges
infra/	Terraform (azurearm)

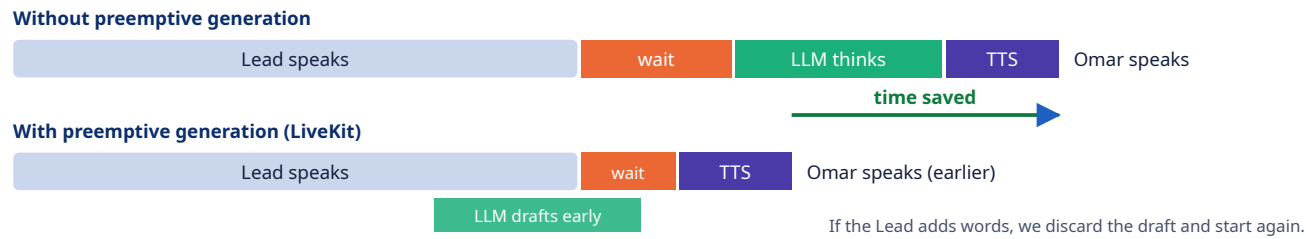
Latency is the time from the end of the Lead's speech to the start of Omar's voice. People reply in about 0.2–0.5 s. Slow replies feel robotic and reduce bookings.

6 Where the time goes (median estimate, ms)



Sources: research tickets 03 and 25. These are estimates, not measurements. Ticket 23 measures the LLM step. Real test calls recalibrate all steps.

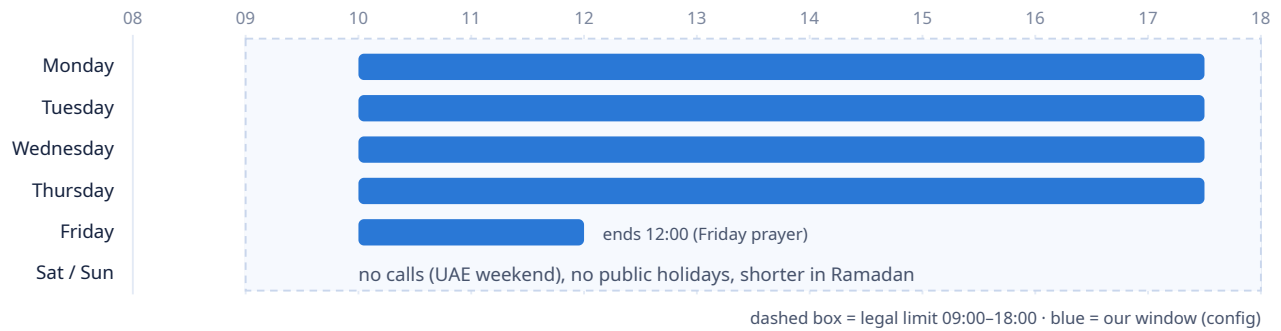
6.1 How we cut the time



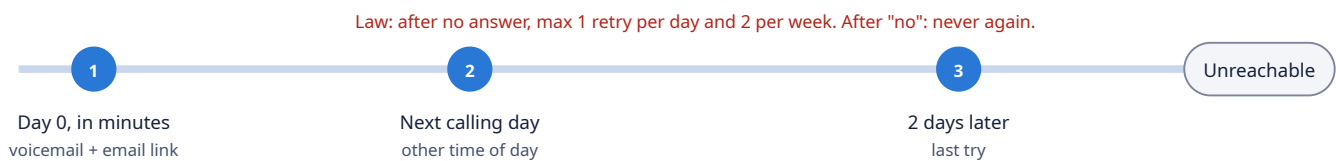
Technique	What it does	Where
Own GPU next to the agent	No network and no queue for the LLM. First token in about 100 ms.	Azure GPU VM
Turn-detection model	Reads the words. Knows "finished" from "pause". Allows a 0.3 s wait, not 0.5 s.	Deepgram Flux (default) or LiveKit model
Preemptive generation	The LLM starts before the turn ends. Optional: preemptive TTS.	LiveKit setting
Stream by sentence	TTS starts on the first sentence. The LLM writes the rest.	LiveKit pipeline
Warm connections, prompt cache	No reconnect for each turn. The fixed prompt part is cached.	Agent worker
Calendar prefetch	Get free slots when Intent to Buy appears. Say a short line while booking.	Tools

Targets (start values): median turn gap ≤ 1.2 s and p95 ≤ 2.0 s as hard limits. The design goal is 0.6–0.8 s. Real calls set the final values.

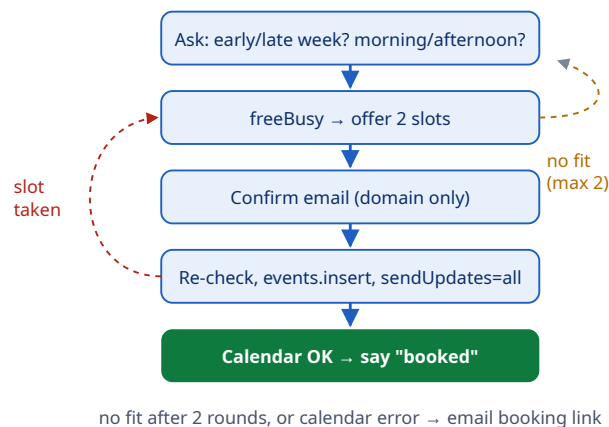
7 When Omar may call (UAE time)



7.1 Call Attempt sequence (no answer)



7.2 Booking a Discovery Call

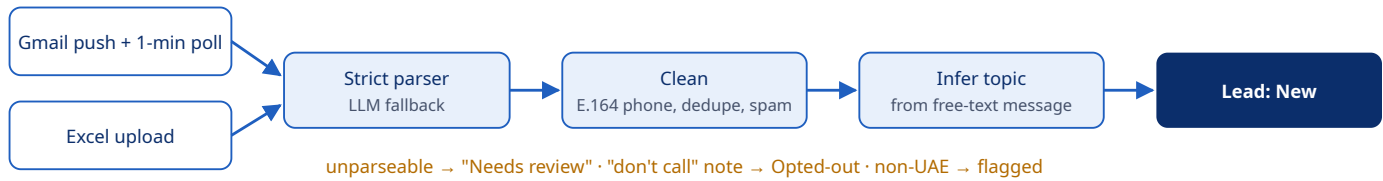


Rule	Value (config)
Length	30 min, Google Meet
Bookable time	Mon–Fri 10:00–18:00
Buffer	15 min around events
Window	2 h to 7 days ahead
Slots per offer	2 (never more)
Fallback	Email link (Google booking page)
Auth	Workspace Internal OAuth app

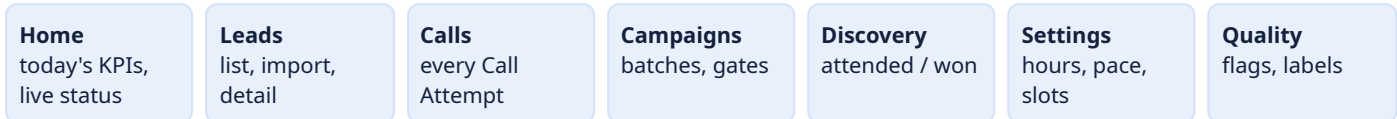
The invite shows only the title, the Meet link and a 3-line agenda. The Manager's private brief goes to the dashboard and to email. It never goes in the invite.

Callbacks: Omar agrees only to times in the calling window. Max 2 callbacks for each Lead.

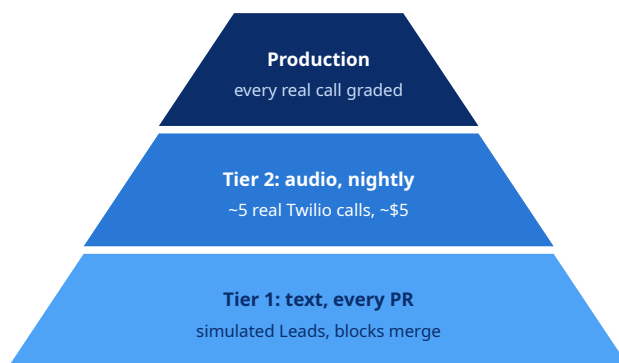
8 Ingestion



Dashboard: 7 screens (Google sign-in, admin only)



9 Evals: three layers



Order of checks

1. Code checks first: end status, legal order, no numbers, no false "booked"
2. Jev judges: one yes/no per failure mode (needy tone, talks over Lead, off-sheet info, missed intent)
3. Unsure verdict → escalate to Claude Sonnet 5
4. Calibrate: Manager labels ~50 real calls
5. Simulated Lead = Claude Haiku 4.5 (other model family than Omar)
6. Each real failure → new Tier 1 scenario

Observability

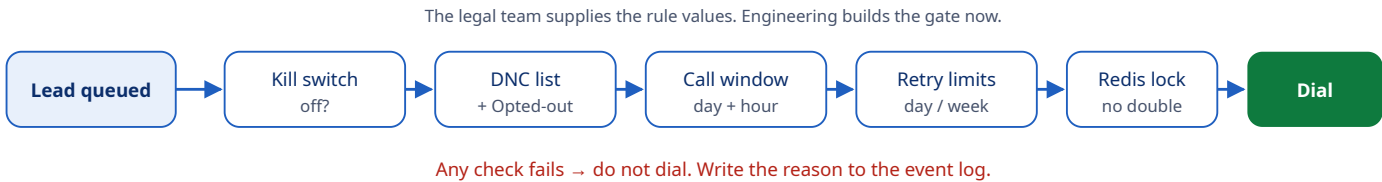
- OpenTelemetry → Langfuse (self-hosted in Azure). One trace for each call, keyed on the call ID. Audio is attached.
- Spans for each turn: end-of-turn, LLM first token, TTS first audio, tool calls.
- Dashboard KPIs: connect rate, booking rate, show-up rate, cost for each Discovery Call, latency p50/p95.

10 When a part fails during a call

Part that fails	What the Lead hears and what happens	Counts as retry?
Own GPU / LLM	Same-turn switch to DeepSeek (EU/US host). Omar can say "one sec..."	No
TTS (Cartesia)	Same-turn switch to ElevenLabs.	No
Calendar	"I will email you a link to pick a time." Lead stays Qualified.	No
STT (Deepgram)	"Sorry, the line is bad. I will call you right back." Callback is scheduled.	Yes (like a dropped line)
All parts	Call ends, callback is scheduled. The dialer pauses until the system is healthy.	Yes

Alerts by email/Slack: GPU down, repeated failovers, dialer self-pause.

10.1 Pre-dial gate (each Call Attempt must pass)



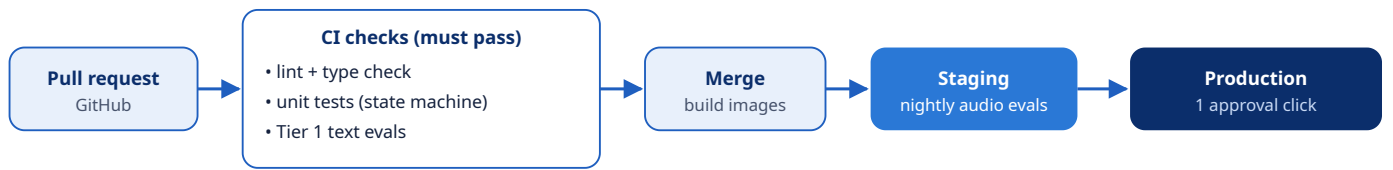
Security and abuse

- Omar has only 4 tools: check calendar, book slot, schedule callback, set outcome. No tool can leak data.
- Omar never tells his instructions or data about other Leads.
- Prompt injection: Omar laughs lightly, says "Nice try!", and goes back to the topic. This is an eval scenario.
- Maximum call length: 10 minutes.
- Phone and email are masked in logs. Data is encrypted at rest.
- Admin kill switch: "Stop all calls now".

Compliance hooks we build now

- Pre-dial gate (above).
- Call register: CSV export from the event log.
- Data requests: "Erase this Lead" and "Export this Lead's data".
- Consent evidence fields: timestamp, IP, form version. The website team adds the checkbox.
- Recording notice in each opening. Call ends if the Lead refuses recording.

11 Environments and delivery



Local: your laptop, LiveKit web playground, Twilio calls to own phones.
Deploys: LiveKit agent workers drain. Live calls finish first. No call drops on a deploy.

11.1 Configuration: where each setting lives

Kind	Examples	Where
Agent behaviour	Prompts, persona, knowledge sheet	Git, gated by evals
Operations	Calling hours, pace, concurrency, slot rules	Database, edited in dashboard, logged
Secrets	API keys, OAuth refresh token	Azure Key Vault
Infrastructure	VMs, Container Apps, DB, Redis, Blob	Terraform in <code>infra/</code>

11.2 Cost for each minute (estimate)

Estimated costs. Each cost centre has a 30% safety margin. We will report the real costs after the first test calls (milestone M2).

Usage cost for each connected minute (USD)

Phone (Twilio, UAE)

0.39

LiveKit (agent + SIP)

0.02

Speak (Cartesia)

0.02

Listen (Deepgram)

0.01

Think (own GPU)

in fixed cost

Usage total ≈ **\$0.44** for each minute

Fixed cost (estimate)	Each month
GPU VM, calling hours only	\$360–550
Azure platform (apps, DB, Redis, storage, Langfuse)	\$200–330
LiveKit Cloud plan	\$65
Nightly audio evals	\$200
Total fixed	\$825–1,145

All-in cost for each minute (usage + share of fixed cost)

Call minutes each month	Share of fixed cost	All-in for each minute	One 3-minute call
1,500 (expected start)	\$0.55–0.76	\$1.00–1.20	\$3.00–3.60
5,000	\$0.17–0.23	\$0.61–0.67	\$1.83–2.01
10,000	\$0.08–0.11	\$0.52–0.56	\$1.56–1.68

Cost levers: if a hosted LLM (for example Groq) is fast enough in ticket 23, we can remove the GPU VM. A local e&du line (v2) can lower the phone cost.

12 Risk map (likelihood × impact, our judgement)

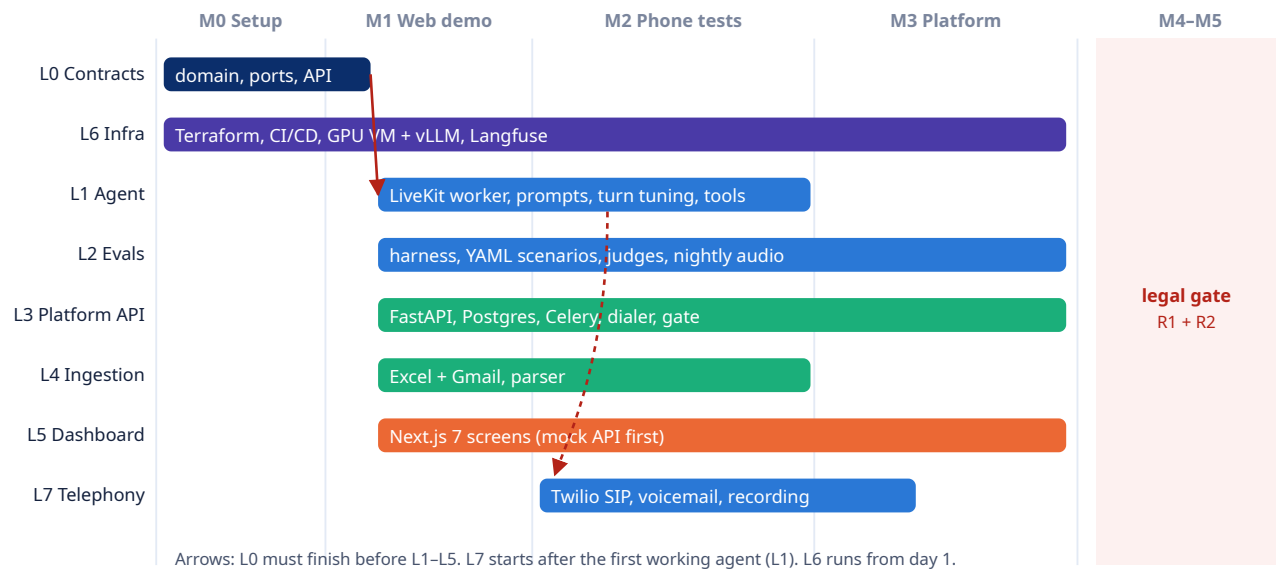


red blocks go-live · amber open or accepted · blue reduced by design. Full register: `.scratch/calling-agent/risks.md`.

ID	Risk	Owner	Action
R1	Foreign caller number (Res 56/2024 Art. 4)	Legal team	e&/du number via Twilio BYOC or direct LiveKit trunk
R2	No telemarketing approval	Legal team	Apply to the competent authority
R3	Backlog Leads without consent box	Owner (accepted)	Show to C-suite; new form gets checkbox
R4	DNC register check not defined	Legal + eng	Gate ready; plug in the official check
R10	LLM data to China	Engineering	Never use DeepSeek's own API; EU/US hosts only
T8	LLM latency not proven	Engineering	Benchmark ticket 23 before lock-in

One engineer builds the system with subagents. We maximise parallel work. Rule: **contracts first** (lane L0). Then most lanes run at the same time.

13 Lanes and dependencies



13.1 Milestones and exit criteria

Milestone	Done when	Size
M0 Setup	Repo, Azure + GPU quota, Twilio/Deepgram/Cartesia/LiveKit accounts, Terraform skeleton	S
M1 Web demo	Talk to Omar in a browser. Legal opening, rapport, objections, fake booking. Tier 1 evals run.	M
M2 Phone tests	Calls to own team, real bookings, recording, voicemail detection. LLM benchmark and voice pick done.	M
M3 Platform	Ingestion, Lead lifecycle, dialer + gate, 7 screens, Langfuse, judges, nightly audio evals	L
M4 Soft launch	After R1 + R2. 10 Leads (all reviewed) → 25 → judges calibrated	S
M5 Full use	Backlog at full pace. New Leads called in minutes.	S

14 Open items

Item	What is needed	Owner
Benchmark LLM hosts (ticket 23)	Own GPU vs Groq (Helsinki) vs Nebius vs DeepSeek. Needs Azure GPU quota and keys.	Engineering
Pick Omar's voice (ticket 24)	4–6 male voices at phone quality. Owner picks by ear.	Owner
Approval, DNC, recording consent (ticket 18)	Rules from the authority. Plug into the gate.	Legal team
e&/du SIP trunk (ticket 22)	Local number on trade licence. Path: Twilio BYOC or direct LiveKit trunk.	Legal team
Website form change (ticket 20)	Consent checkbox, Service dropdown, evidence fields.	Website team

14.1 File map

Decisions (read first)

CONTEXT.md · glossary

docs/adr/0001 · state machine

docs/adr/0002 · LiveKit

.scratch/calling-agent/map.md

index of all decisions

.scratch/.../issues/NN-*.md

one ticket per decision

.scratch/.../risks.md

C-suite risk register

Evidence

research/01 · UAE law

research/02 · Twilio in UAE

research/03 · STT/TTS/LLM

research/04 · Google Calendar

research/05 · evals practice

research/25 · all-UAE stack

assets/agency-knowledge.md

what Omar may say

assets/dummy-leads/

test data with mess

Try it

prototypes/

PROTOTYPE-call-flow.html

click through 10 call scenarios

assets/accent-clips/SCRIPT.md

for a later STT accent test

docs/pdf-src/

source of these PDFs

Rules for new engineers. Use the glossary words. Put each new decision in a ticket or an ADR. Do not put business logic in Next.js. Do not let the LLM own a legal rule. Add a test scenario for each real failure.